

Categories and Labels

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sound Becomes a Signal

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

AI or Not?

Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Replace a vague instruction for Categories and operational labels. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Categories and operational labels. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Transfer the method to a new record

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y. Label revise if any known field fails; otherwise label review if a field is blank or unreadable; otherwise label ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code. Explain whether ready alone grants permission to borrow.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Your dataset labels 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational label rule and classify this boundary case under that rule. Explain why a different rule could be defensible.

6. Solve the given case

Six cards show two circles, two triangles and two squares. Sort by shape.

Why can red and blue circles share a label?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

The rule is: fruit has seeds; other objects go in other. Sort an apple, a pencil and a sealed mystery box. Which item needs more evidence, and why?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 7 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Define the Classroom Sorter's bin labels and write how to handle objects that reasonably fit more than one bin.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Your dataset labels 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational label rule and classify this boundary case under that rule. Explain why a different rule could be defensible.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Apply the calendar guide

The calendar policy accepts whole-number days 1 through 30 inclusive and a room code A or B. Check known failures first, then unknown fields, then readiness. Label these cards and give evidence: day 30 room B; day 31 room A; day 14 room unreadable; day 31 room unreadable.

My reasoning and evidence

13. W2 Replace a vague instruction

A helper says Label a card ready if it looks nice. Explain why this can lead to inconsistent labels. Replace it with an instruction that another person can apply using the calendar policy, including what to do about a blank field.

My reasoning and evidence

14. W3 Count and investigate disagreement

For the same four cards, Mina records ready, revise, review, ready. Theo records ready, review, review, revise. How many positions agree? Identify the disagreements. Can you decide which annotator is correct from these label lists alone? Explain what else you need.

My reasoning and evidence

15. W4 Transfer the method to a new record

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y. Label revise if any known field fails; otherwise label review if a field is blank or unreadable; otherwise label ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code. Explain whether ready alone grants permission to borrow.

My reasoning and evidence

16. W5 Keep a policy change visible

Version 1 accepts calendar days 1 through 30. Version 2 accepts days 1 through 31. A card says day 31 room A. State its label under each version. Explain why mixing the two labels without recording the version could create a misleading training dataset.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Categories and Labels

A fictional craft club labels cardboard-order cards ready, review or revise.

Allowed quantity: whole number 1–4. Required colour: red or blue.

If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards

A=(4,red), B=(5,blank), C=(2,blank).

AI ACADEMY / AI-1/7

Categories and Labels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read policy

Check known invalid fields before checking missing ones.



2. Card A

4 is allowed and red is allowed: ready.



3. Card B

5 is outside 1-4: revise, even though colour is missing.



4. Card C

2 is valid but colour is blank: review.

CHANGE ONE THING

Change B's quantity from 5 to 4; leave colour blank.

Explain the changed result using the matching lesson.

Explain why B originally was revise rather than review, then label its changed version.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Write a sorter label guide with examples of complete, invalid and unresolved inputs before comparing labels.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Write a sorter label guide with examples of complete, invalid and unresolved inputs before comparing labels.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.
